

Databases I – Exercises

Semester 3 (Theory)

Mathis Neunzig

DHBW Mosbach

Autumn 2026

Table of Contents

1 Relational Databases – Keys

- Exercise 1A – University Schema
- Exercise 1B – Airline Schema
- Exercise 1C – Event Schema
- Exercise 1D – Music Schema

2 Relational Algebra

- Exercise 2A – Customer / Supplier / Good (Basic)
- Exercise 2B – Customer / Supplier / Good (Harder)
- Exercise 2C – PC / Laptop / Printer (Intermediate)
- Exercise 2D – PC / Laptop / Printer (Advanced Algebra)

3 Normal Forms (1NF – BCNF)

- Exercise 3A – Eurovision Song Contest (Worked Example)
- Exercise 3B – Ticket Schema
- Exercise 3C – Library Loan Schema

4 Normal Forms (4NF / 5NF)

- Exercise 4A – Train / Conductor / Mechanic (4NF)
- Exercise 4B – Line / Station / Day (5NF)
- Exercise 4C – Train / Route / Service / CarriageType (4NF)
- Exercise 4D – Line / Driver / Shift / Depot (5NF)

5 Canonical Cover & Synthesis Algorithm

Table of Contents

1 Relational Databases – Keys

- Exercise 1A – University Schema
- Exercise 1B – Airline Schema
- Exercise 1C – Event Schema
- Exercise 1D – Music Schema

2 Relational Algebra

- Exercise 2A – Customer / Supplier / Good (Basic)
- Exercise 2B – Customer / Supplier / Good (Harder)
- Exercise 2C – PC / Laptop / Printer (Intermediate)
- Exercise 2D – PC / Laptop / Printer (Advanced Algebra)

3 Normal Forms (1NF – BCNF)

- Exercise 3A – Eurovision Song Contest (Worked Example)
- Exercise 3B – Ticket Schema
- Exercise 3C – Library Loan Schema

4 Normal Forms (4NF / 5NF)

- Exercise 4A – Train / Conductor / Mechanic (4NF)
- Exercise 4B – Line / Station / Day (5NF)
- Exercise 4C – Train / Route / Service / CarriageType (4NF)
- Exercise 4D – Line / Driver / Shift / Depot (5NF)

5 Canonical Cover & Synthesis Algorithm

Exercise 1A – University Schema

Relational Schema

- **Student**(MatrNr, FirstName, LastName, Email, DOB, MajorID^{FK})
- **Department**(DeptID, Name, Building, HeadProfID^{FK})
- **Course**(CourselD, Title, Credits, DeptID^{FK})
- **Enrollment**(MatrNr^{FK}, CourselD^{FK}, Semester, Grade)
- **Professor**(ProfID, FirstName, LastName, Email, DeptID^{FK})

Tasks – answer for every relation

- 1 List all **attributes** and their domains.
- 2 Identify all **candidate keys**; select and justify your **primary key**.
- 3 Identify all **foreign keys** and state which PK each one references.
- 4 Classify every key as *simple/composite* and *natural/surrogate*.
- 5 Create **2 realistic sample tuples** for each relation.

Difficulty: Simple Time: 15 min

Exercise 1B – Airline Schema

Relational Schema

- **Airline**(AirlineCode, Name, Country, FoundedYear)
- **Airport**(IATA, Name, City, Country, Terminals)
- **Connection**(ConnectionID, DepartureIATA^{FK}, ArrivalIATA^{FK}, AirlineCode^{FK}, FlightNumber)
- **Schedule**(ConnectionID^{FK}, Date, DepartureTime, ArrivalTime, Status)
- **Passenger**(PassengerID, FirstName, LastName, Passport, Nationality)

Tasks – same 5 questions as 1A

- 1 List all **attributes** and their domains.
- 2 Identify all **candidate keys**; select and justify your **primary key**.
- 3 Identify all **foreign keys** and state which PK each one references.
- 4 Classify every key as *simple/composite* and *natural/surrogate*.
- 5 Create **2 realistic sample tuples** for each relation.

Difficulty: Simple Time: 15 min

Exercise 1C – Event Schema

Relational Schema

- **Event**(EventID, Title, Date, Location, OrganizerID^{FK})
- **Organizer**(OrgID, Name, Email, Phone)
- **Sponsor**(SponsorID, CompanyName, ContactEmail, Budget)
- **Sponsorship**(EventID^{FK}, SponsorID^{FK}, Amount, ContractDate)
- **Attendee**(AttendeeID, FirstName, LastName, Email)
- **Attendance**(AttendeeID^{FK}, EventID^{FK}, RegistrationDate, Paid)

Tasks – same 5 questions as 1A

- 1 List all **attributes** and their domains.
- 2 Identify all **candidate keys**; select and justify your **primary key**.
- 3 Identify all **foreign keys** and state which PK each one references.
- 4 Classify every key as *simple/composite* and *natural/surrogate*.
- 5 Create **2 realistic sample tuples** for each relation.

Exercise 1D – Music Schema

Relational Schema

- **Artist**(ArtistID, Name, Country, DebutYear, Genre)
- **Album**(AlbumID, Title, ArtistID^{FK}, ReleaseYear, Label)
- **Track**(TrackID, Title, AlbumID^{FK}, TrackNumber, Duration)
- **Features**(TrackID^{FK}, FeaturedArtistID^{FK})
- **Playlist**(PlaylistID, Name, UserID, CreatedDate)
- **PlaylistTrack**(PlaylistID^{FK}, TrackID^{FK}, AddedDate, Position)

Tasks – same 5 questions as 1A

- 1 List all **attributes** and their domains.
- 2 Identify all **candidate keys**; select and justify your **primary key**.
- 3 Identify all **foreign keys** and state which PK each one references.
- 4 Classify every key as *simple/composite* and *natural/surrogate*.
- 5 Create **2 realistic sample tuples** for each relation.

Table of Contents

1 Relational Databases – Keys

- Exercise 1A – University Schema
- Exercise 1B – Airline Schema
- Exercise 1C – Event Schema
- Exercise 1D – Music Schema

2 Relational Algebra

- Exercise 2A – Customer / Supplier / Good (Basic)
- Exercise 2B – Customer / Supplier / Good (Harder)
- Exercise 2C – PC / Laptop / Printer (Intermediate)
- Exercise 2D – PC / Laptop / Printer (Advanced Algebra)

3 Normal Forms (1NF – BCNF)

- Exercise 3A – Eurovision Song Contest (Worked Example)
- Exercise 3B – Ticket Schema
- Exercise 3C – Library Loan Schema

4 Normal Forms (4NF / 5NF)

- Exercise 4A – Train / Conductor / Mechanic (4NF)
- Exercise 4B – Line / Station / Day (5NF)
- Exercise 4C – Train / Route / Service / CarriageType (4NF)
- Exercise 4D – Line / Driver / Shift / Depot (5NF)

5 Canonical Cover & Synthesis Algorithm

Exercise 2A – Schema

Schema

- **Customer**(CustID, Name, City, Balance)
- **Supplier**(SupplierID, Name, City, ContactEmail)
- **Good**(GoodID, Name, Price, SupplierID^{FK})

CustID	Name	City	Balance
C1	Maier GmbH	Mosbach	240.00
C2	Schulz AG	Heidelberg	-50.00

SupplierID	Name	City
S1	Fresh Foods	Mosbach
S2	Top Drinks	Heidelberg

GoodID	Name	Price	SupplierID
G1	Apple	0.49	S1
G2	Orange Juice	2.99	S2
G3	Sparkling Water	6.50	S1

Exercise 2A – Tasks

Tasks – express in relational algebra

- 1 List all goods with price > 5.00 (show Name, Price).
- 2 List all suppliers located in Mosbach.
- 3 List all customers with a **negative** balance.
- 4 Find all goods supplied by suppliers in Mosbach.
- 5 List customer names and cities only (project exactly these two columns).
- 6 Find customers whose balance is less than the price of the cheapest good.

Reminder: operators

$\sigma_{\text{condition}}(R)$ $\pi_{\text{attrs}}(R)$ $R_1 \bowtie R_2$ $R_1 \times R_2$ $R_1 \setminus R_2$

Difficulty: Simple Time: 15 min

Exercise 2B – Tasks

Schema (same as 2A, Supplier now also has a Cost column)

- **Customer**(CustID, Name, City, Balance)
- **Supplier**(SupplierID, Name, City, ContactEmail, Cost)
- **Good**(GoodID, Name, Price, SupplierID^{FK})

Tasks

- 1 Find goods where the price margin ($\text{Price} - \text{Cost}$) > 2.00 .
- 2 List all suppliers in Heidelberg.
- 3 List customers whose balance exceeds the price of the most expensive good.
- 4 Find all goods **not** supplied by suppliers in Mosbach.
- 5 Find all customer-supplier pairs from the **same city** (use Cartesian product + restriction).

Difficulty: Simple–Intermediate Time: 20 min

Exercise 2C – Schema

Schema

- **Product**(ProductID, Maker, Type)
- **PC**(ProductID^{FK}, Speed_GHz, RAM_GB, HDD_GB, Price)
- **Laptop**(ProductID^{FK}, Speed_GHz, RAM_GB, HDD_GB, Screen_inch, Price)
- **Printer**(ProductID^{FK}, Color, Type, Price)

ProductID	Maker	Type
P01	Dell	PC
P02	Dell	Laptop
P03	HP	PC
P04	HP	Printer
P05	Apple	Laptop

ProductID	GHz	RAM	HDD	Price
P01	3.2	16	512	899.00
P03	2.8	8	256	649.00

Exercise 2C – Tasks

Tasks

- 1 List all PCs with speed > 3.0 GHz.
- 2 Find all manufacturers who make **both** PCs and Laptops.
- 3 List all products from manufacturer 'Dell'.
- 4 Find all PC–Laptop pairs from the **same manufacturer** (use Cartesian product).
- 5 Find makers who produce at least **3 different types** of products.

Hint for task 2

Use projection + intersection:

$$\pi_{\text{Maker}}(\sigma_{\text{Type}='PC'}(\text{Product})) \cap \pi_{\text{Maker}}(\sigma_{\text{Type}='Laptop'}(\text{Product}))$$

Difficulty: Intermediate Time: 20 min

Exercise 2D – Tasks

Schema (same as 2C)

Product(ProductID, Maker, Type) **PC**(ProductID, Speed_GHz, RAM_GB, HDD_GB, Price)

Laptop(ProductID, Speed_GHz, RAM_GB, HDD_GB, Screen_inch, Price)

Printer(ProductID, Color, Type, Price)

Tasks

- 1 Find laptops with screen size ≥ 15 inch **and** price < 1000 .
- 2 List all manufacturers who **only** make printers (not PCs or Laptops).
- 3 Find all products more expensive than the **average** PC price (use γ).
- 4 Count products per manufacturer using γ (grouping/aggregation).
- 5 Write as relational algebra: “Find makers of PCs with speed > 3 GHz” – express as a single nested expression.

Difficulty: Intermediate Time: 25 min

Table of Contents

1 Relational Databases – Keys

- Exercise 1A – University Schema
- Exercise 1B – Airline Schema
- Exercise 1C – Event Schema
- Exercise 1D – Music Schema

2 Relational Algebra

- Exercise 2A – Customer / Supplier / Good (Basic)
- Exercise 2B – Customer / Supplier / Good (Harder)
- Exercise 2C – PC / Laptop / Printer (Intermediate)
- Exercise 2D – PC / Laptop / Printer (Advanced Algebra)

3 Normal Forms (1NF – BCNF)

- Exercise 3A – Eurovision Song Contest (Worked Example)
- Exercise 3B – Ticket Schema
- Exercise 3C – Library Loan Schema

4 Normal Forms (4NF / 5NF)

- Exercise 4A – Train / Conductor / Mechanic (4NF)
- Exercise 4B – Line / Station / Day (5NF)
- Exercise 4C – Train / Route / Service / CarriageType (4NF)
- Exercise 4D – Line / Driver / Shift / Depot (5NF)

5 Canonical Cover & Synthesis Algorithm

Exercise 3A – Eurovision Song Contest

Starting Relation

$R(\text{Year}, \text{Country}, \text{Song}, \text{Artist}, \text{Points}, \text{Rank}, \text{Host_Country}, \text{Host_City})$

Year	Country	Song	Artist	Pts	Rank	Host	City
2023	Sweden	Tattoo	Loreen	583	1	UK	Liverpool
2023	Finland	Cha Cha Cha	Käärijä	526	2	UK	Liverpool
2022	Ukraine	Stefania	Kalush Orchestra	631	1	Italy	Turin
2022	UK	Space Man	Sam Ryder	466	2	Italy	Turin

Tasks

- 1 Identify all violations of 1NF, 2NF, 3NF, BCNF.
- 2 Decompose to **1NF** (if needed).
- 3 Decompose to **2NF** – show functional dependencies, show decomposed tables.
- 4 Decompose to **3NF** – identify and eliminate transitive FDs.
- 5 Check if **BCNF** is achieved; decompose further if necessary.

Exercise 3A – Functional Dependencies

Identify the Functional Dependencies

Working from the relation

$R(\text{Year}, \text{Country}, \text{Song}, \text{Artist}, \text{Points}, \text{Rank}, \text{Host_Country}, \text{Host_City})$:

- Which attributes uniquely identify a tuple?
- What does $\{\text{Year}, \text{Country}\}$ determine?
- What does Year alone determine?
- Is there a transitive dependency (e.g. $A \rightarrow B \rightarrow C$)?

Guiding Questions for Each Normal Form

1NF Are all attribute values atomic? Any repeating groups?

2NF Are there non-prime attributes that depend on only *part* of the PK?

3NF Are there non-prime attributes that transitively depend on the PK?

BCNF Does every non-trivial FD have a **superkey** on the left?

Difficulty: Simple–Intermediate Time: 25 min

Exercise 3B – Ticket Schema

Starting Relation

Ticket(TicketID, EventName, EventDate, Venue, VenueCity, CustomerName, CustomerEmail, SeatNumber, Price, TicketType)

Known Functional Dependencies

- TicketID $\rightarrow$ all other attributes
- EventName $\rightarrow$ EventDate, Venue
- Venue $\rightarrow$ VenueCity
- CustomerEmail $\rightarrow$ CustomerName
- {TicketID, SeatNumber} $\rightarrow$ Price (if modelled as composite)

Tasks

- 1 Identify all violations of 1NF, 2NF, 3NF, BCNF.
- 2 Decompose step by step to 1NF $\rightarrow$ 2NF $\rightarrow$ 3NF.
- 3 Show all intermediate tables after each decomposition step.
- 4 Check if BCNF is achieved.

Exercise 3C – Library Loan Schema

Starting Relation

LibraryLoan(LoanID, BookISBN, BookTitle, AuthorName, BorrowerID, BorrowerName, BorrowerEmail, LoanDate, ReturnDate, LibraryBranch, BranchCity)

Known Functional Dependencies

- LoanID $\rightarrow$ all other attributes
- BookISBN $\rightarrow$ BookTitle, AuthorName
- BorrowerID $\rightarrow$ BorrowerName, BorrowerEmail
- LibraryBranch $\rightarrow$ BranchCity

Tasks

- 1 Identify all violations of 1NF, 2NF, 3NF, BCNF.
- 2 Decompose step by step to 1NF $\rightarrow$ 2NF $\rightarrow$ 3NF.
- 3 Show all intermediate tables after each decomposition step.
- 4 Verify lossless-join property after final decomposition.

Table of Contents

1 Relational Databases – Keys

- Exercise 1A – University Schema
- Exercise 1B – Airline Schema
- Exercise 1C – Event Schema
- Exercise 1D – Music Schema

2 Relational Algebra

- Exercise 2A – Customer / Supplier / Good (Basic)
- Exercise 2B – Customer / Supplier / Good (Harder)
- Exercise 2C – PC / Laptop / Printer (Intermediate)
- Exercise 2D – PC / Laptop / Printer (Advanced Algebra)

3 Normal Forms (1NF – BCNF)

- Exercise 3A – Eurovision Song Contest (Worked Example)
- Exercise 3B – Ticket Schema
- Exercise 3C – Library Loan Schema

4 Normal Forms (4NF / 5NF)

- Exercise 4A – Train / Conductor / Mechanic (4NF)
- Exercise 4B – Line / Station / Day (5NF)
- Exercise 4C – Train / Route / Service / CarriageType (4NF)
- Exercise 4D – Line / Driver / Shift / Depot (5NF)

5 Canonical Cover & Synthesis Algorithm

Exercise 4A – Train / Conductor / Mechanic

Starting Relation

$R(\text{Train, Conductor, Mechanic})$

Multi-Valued Dependencies (MVDs)

- $\text{Train} \twoheadrightarrow \text{Conductor}$ (a train can be driven by multiple conductors, independently of mechanics)
- $\text{Train} \twoheadrightarrow \text{Mechanic}$ (a train can be maintained by multiple mechanics, independently)

Train	Conductor	Mechanic
ICE 42	Hoffmann	Bauer
ICE 42	Hoffmann	Klein
ICE 42	Neumann	Bauer
ICE 42	Neumann	Klein

Tasks

- 1 Identify the MVDs and explain the resulting redundancy.
- 2 Decompose to 4NF – show the decomposed tables.

Exercise 4B – Line / Station / Day

Starting Relation

$R(\text{Line}, \text{Station}, \text{Day})$

Business Constraint (Join Dependency)

If line L stops at station S , and line L runs on day D , and station S is served on day D – then the combination (L, S, D) **must** exist in R .

Line	Station	Day
RE7	Mosbach	Monday
RE7	Heidelberg	Monday
RE7	Mosbach	Friday
RE7	Heidelberg	Friday

Tasks

- 1 Identify the **join dependency** and explain why 4NF is not sufficient.
- 2 Decompose to **5NF** – show the three binary projections.
- 3 Reconstruct the original relation by joining the projections and verify no information is lost.

Exercise 4C – Train / Route / Service / CarriageType

Starting Relation

$R(\text{Train}, \text{Route}, \text{Service}, \text{CarriageType})$

Multi-Valued Dependencies

All three MVDs hold independently:

- $\text{Train} \twoheadrightarrow \text{Route}$
- $\text{Train} \twoheadrightarrow \text{Service}$
- $\text{Train} \twoheadrightarrow \text{CarriageType}$

(A train serves multiple routes, offers multiple services, and can have multiple carriage types – all choices are independent of each other.)

Tasks

- 1 Explain why storing all combinations leads to redundancy.
- 2 Decompose to **4NF** by eliminating all non-trivial MVDs.
- 3 How many tables result? Write out each table schema.
- 4 Verify lossless-join property.

Exercise 4D – Line / Driver / Shift / Depot

Starting Relation

$R(\text{Line}, \text{Driver}, \text{Shift}, \text{Depot})$

Business Rules

- Each driver can work on multiple lines.
- Each line operates in multiple shifts.
- Drivers are assigned to depots.
- The combination (L, D, S, Dp) is valid if and only if each pairwise assignment is valid **and** the combination is consistent.

This results in a **join dependency** $\bowtie \{(L, D), (L, S), (D, \text{Dp})\}$.

Tasks

- 1 Identify the join dependency.
- 2 Decompose to **5NF** – state each binary/ternary relation.
- 3 Demonstrate that the original relation can be reconstructed by natural join.

Table of Contents

1 Relational Databases – Keys

- Exercise 1A – University Schema
- Exercise 1B – Airline Schema
- Exercise 1C – Event Schema
- Exercise 1D – Music Schema

2 Relational Algebra

- Exercise 2A – Customer / Supplier / Good (Basic)
- Exercise 2B – Customer / Supplier / Good (Harder)
- Exercise 2C – PC / Laptop / Printer (Intermediate)
- Exercise 2D – PC / Laptop / Printer (Advanced Algebra)

3 Normal Forms (1NF – BCNF)

- Exercise 3A – Eurovision Song Contest (Worked Example)
- Exercise 3B – Ticket Schema
- Exercise 3C – Library Loan Schema

4 Normal Forms (4NF / 5NF)

- Exercise 4A – Train / Conductor / Mechanic (4NF)
- Exercise 4B – Line / Station / Day (5NF)
- Exercise 4C – Train / Route / Service / CarriageType (4NF)
- Exercise 4D – Line / Driver / Shift / Depot (5NF)

5 Canonical Cover & Synthesis Algorithm

Exercise 5A – Canonical Cover & Synthesis

Given

$$R = \{A, B, C, D\}$$

$$F = \{D \rightarrow AB, \quad CD \rightarrow B, \quad C \rightarrow B\}$$

Tasks

- ① Compute the **canonical cover** F_c :
 - Decompose right-hand sides into single attributes.
 - Eliminate extraneous attributes from left-hand sides.
 - Eliminate redundant FDs.
- ② Apply the **Synthesis Algorithm** to obtain a **3NF** decomposition.
- ③ Check whether the result is:
 - **Dependency-preserving**: can all FDs in F be checked within a single table?
 - **Lossless-join**: does the natural join of all tables reconstruct R ?

Difficulty: Intermediate Time: 25 min

Exercise 5B – Canonical Cover & Synthesis

Given

$$R = \{A, B, C, D, E\}$$

$$F = \{CD \rightarrow A, \quad D \rightarrow AE, \quad C \rightarrow BE, \quad ABD \rightarrow CE, \quad E \rightarrow A\}$$

Tasks

- 1 Compute the **canonical cover** F_c .
- 2 Determine all **candidate keys** of R .
- 3 Apply the **Synthesis Algorithm** to obtain a **3NF** decomposition.
- 4 Check whether the result is dependency-preserving and lossless-join.

Reminder: Attribute Closure

To check if X is a superkey: compute X^+ under F .

If $X^+ = R$, then X is a superkey.

Difficulty: Intermediate–Advanced **Time:** 30 min

Exercise 5C – Canonical Cover & Synthesis

Given

$$R = \{A, B, C, D, E, F\}$$

$$F = \{C \rightarrow DE, \quad ABC \rightarrow DF, \quad ABF \rightarrow CDE, \quad ABD \rightarrow CF, \quad CEF \rightarrow D\}$$

Tasks

- 1 Compute the **canonical cover** F_c .
- 2 Determine all **candidate keys** of R .
- 3 Apply the **Synthesis Algorithm** to obtain a **3NF** decomposition.
- 4 Check dependency-preservation and lossless-join.
- 5 Optional: attempt **BCNF** decomposition and compare the result.

Difficulty: Advanced Time: 35 min

Table of Contents

1 Relational Databases – Keys

- Exercise 1A – University Schema
- Exercise 1B – Airline Schema
- Exercise 1C – Event Schema
- Exercise 1D – Music Schema

2 Relational Algebra

- Exercise 2A – Customer / Supplier / Good (Basic)
- Exercise 2B – Customer / Supplier / Good (Harder)
- Exercise 2C – PC / Laptop / Printer (Intermediate)
- Exercise 2D – PC / Laptop / Printer (Advanced Algebra)

3 Normal Forms (1NF – BCNF)

- Exercise 3A – Eurovision Song Contest (Worked Example)
- Exercise 3B – Ticket Schema
- Exercise 3C – Library Loan Schema

4 Normal Forms (4NF / 5NF)

- Exercise 4A – Train / Conductor / Mechanic (4NF)
- Exercise 4B – Line / Station / Day (5NF)
- Exercise 4C – Train / Route / Service / CarriageType (4NF)
- Exercise 4D – Line / Driver / Shift / Depot (5NF)

5 Canonical Cover & Synthesis Algorithm

Exercise 6A – Airplane / Flight / Pilot

Description

- An **Airplane** has: registration number, type, and capacity.
- A **Flight** has: flight number, departure airport, arrival airport, and date.
- A **Pilot** has: staff number, name, and license type.
- Each flight uses **exactly one** airplane.
- Each flight is operated by **exactly one** pilot.
- A pilot can operate **many** flights; an airplane can be used for **many** flights.

Task

Draw the **Entity-Relationship Diagram** in Chen's notation.

- Show all entities with their attributes (underline key attributes).
- Show all relationships with their names.
- Annotate all cardinalities (1, N, M notation or min-max).

Difficulty: Simple Time: 15 min

Exercise 6B – Detailed Flight Schema

Description

- A **Flight** (identified by FlightNumber) consists of multiple **FlightLegs**.
- A **FlightLeg** connects one **Airport** to another (origin and destination).
- Each **Airport** has: IATA code, name, and city.
- A **DailyFlightLeg** is a scheduled instance of a FlightLeg on a specific date – it is a **weak entity**.
- An **AircraftType** has: a name and seat configuration.
- Each DailyFlightLeg uses exactly one AircraftType.

Task

Draw the complete **Extended ERM** in Chen's notation.

- Identify and model **weak entities** with double rectangles and double diamonds.
- Annotate all cardinalities.
- Indicate all identifying relationships.

Exercise 6C – Crew with Specialization

Description

- **Crew members** are either **Pilots** or **Flight Attendants** (total, disjoint specialization).
- All crew members share: staff number, name, *languages spoken* (multivalued).
- **Pilots** additionally have: license number, license type, and a list of aircraft ratings (multivalued – aircraft types they are certified for).
- **Flight Attendants** additionally have: seniority level.
- All crew members receive **emergency training**; each training record has a date and type.

Task

Draw the **Extended ERM** using generalization/specialization.

- Use the ISA triangle notation with disjoint/total markers.
- Show all attributes, including multivalued attributes (double ellipses).
- Show the `EmergencyTraining` relationship with its attributes.

Table of Contents

1 Relational Databases – Keys

- Exercise 1A – University Schema
- Exercise 1B – Airline Schema
- Exercise 1C – Event Schema
- Exercise 1D – Music Schema

2 Relational Algebra

- Exercise 2A – Customer / Supplier / Good (Basic)
- Exercise 2B – Customer / Supplier / Good (Harder)
- Exercise 2C – PC / Laptop / Printer (Intermediate)
- Exercise 2D – PC / Laptop / Printer (Advanced Algebra)

3 Normal Forms (1NF – BCNF)

- Exercise 3A – Eurovision Song Contest (Worked Example)
- Exercise 3B – Ticket Schema
- Exercise 3C – Library Loan Schema

4 Normal Forms (4NF / 5NF)

- Exercise 4A – Train / Conductor / Mechanic (4NF)
- Exercise 4B – Line / Station / Day (5NF)
- Exercise 4C – Train / Route / Service / CarriageType (4NF)
- Exercise 4D – Line / Driver / Shift / Depot (5NF)

5 Canonical Cover & Synthesis Algorithm

Description

- A **Book** has: title, ISBN, and publication year.
- Each Book belongs to exactly one **Publisher**; a Publisher has: name and address.
- Each Book has multiple **Chapters** – a Chapter is a **weak entity** (exists only as part of a Book); it has a title and page numbers.
- **Authors** write Books (m:n); an Author can write multiple books; a Book can have multiple authors.
- An Author has: name and a list of previous publications (**multivalued attribute**).

Task

Draw the **Extended ERM**.

- Model the weak entity with double rectangle and double diamond.
- Model the multivalued attribute with double ellipse.
- Show the m:n relationship between Author and Book.

Exercise 7B – Book with Overlapping Specialization

Description

- A **Book** is a weak entity with **Chapter** as an even weaker entity.
- **Contributors** are people who contribute to books; they can be **Authors**, **Editors**, or **both** – this is an **overlapping, partial** specialization.
- **Authors** write Chapters (m:n); a Chapter can be written by multiple Authors.
- **Editors** edit entire Books (m:n).
- An Editor additionally has an editorial certification attribute.

Task

Draw the **Extended ERM**.

- Indicate **overlapping** (“O”) and **partial** (“P”) markers on the ISA triangle.
- Show the “writes” (Author–Chapter) and “edits” (Editor–Book) relationships.
- Annotate all cardinalities.

Difficulty: Intermediate Time: 25 min

Exercise 7C – Complex Publication Schema

Description

- **Publication** is a supertype with subtypes **Book** and **Proceedings** (total, disjoint).
- A **Book** has **Editions** (weak entity); each **Edition** has **Chapters** (weak entity of **Edition**); each **Chapter** has **Sections** (weak entity of **Chapter**).
- A **Proceedings** belongs to a **Conference** (with name, year, location).
- A **Copy** is a physical instance of an **Edition** (weak entity of **Edition**).
- **Contributors** specialize into **Authors** and **Editors** (partial, overlapping).

Task

Draw the complete **Extended ERM**.

- Model all levels of nested weak entities.
- Show both specialization hierarchies.
- Annotate all cardinalities and identifying relationships.

Difficulty: Advanced Time: 35 min

Table of Contents

1 Relational Databases – Keys

- Exercise 1A – University Schema
- Exercise 1B – Airline Schema
- Exercise 1C – Event Schema
- Exercise 1D – Music Schema

2 Relational Algebra

- Exercise 2A – Customer / Supplier / Good (Basic)
- Exercise 2B – Customer / Supplier / Good (Harder)
- Exercise 2C – PC / Laptop / Printer (Intermediate)
- Exercise 2D – PC / Laptop / Printer (Advanced Algebra)

3 Normal Forms (1NF – BCNF)

- Exercise 3A – Eurovision Song Contest (Worked Example)
- Exercise 3B – Ticket Schema
- Exercise 3C – Library Loan Schema

4 Normal Forms (4NF / 5NF)

- Exercise 4A – Train / Conductor / Mechanic (4NF)
- Exercise 4B – Line / Station / Day (5NF)
- Exercise 4C – Train / Route / Service / CarriageType (4NF)
- Exercise 4D – Line / Driver / Shift / Depot (5NF)

5 Canonical Cover & Synthesis Algorithm

Exercise 8A – Customers / Discounts / Items / Categories

Description

- **Customers** can have **Discounts** (1:1 – each customer has at most one active discount; a Discount has percentage and validity period).
- **Items** belong to **Categories** (m:n – an item can be in multiple categories; a Category has name and description).
- An Item has: price, name, and description.

Tasks

- 1 Draw the **EERM** with all entities, relationships, attributes, and cardinalities.
- 2 **Map to a relational schema** applying all standard mapping rules:
 - Strong entities → tables with PK.
 - 1:1 relationship → FK in one of the participating tables.
 - m:n relationship → separate junction table with composite PK.
- 3 Write **SQL CREATE TABLE** statements (DDL) for all tables, including all constraints.

Difficulty: Simple Time: 25 min

Exercise 8B – SAP Sales Document Schema

Description (simplified SAP SD module)

- A **SalesDocument** has a **SalesDocumentType** (standard order, rush order, ...).
- Each SalesDocument goes through a **CreditCheckResult**.
- A SalesDocument produces a **BillingDocument** (**BillingType**: invoice, credit memo).
- Each SalesDocument generates a **DeliveryDocument** (**DeliveryType**: outbound delivery).
- Documents can carry a **Block** (delivery block, billing block, credit block).

Tasks

- 1 Draw the **EERM** with all entities, type attributes, and cardinalities.
- 2 Derive the complete **relational schema** with all FKs.
- 3 Identify which relationships are 1:1, 1:N, or M:N.

Difficulty: Intermediate Time: 30 min

Exercise 8C – SAP Organizational Structure

Entities and Hierarchy

Plant	(PlantCode, Name, Address)
CompanyCode	(CompanyCode, Name, Currency)
StorageLocation	(StorageLocationCode, Name, PlantCode ^{FK})
PurchasingOrganization	(PurchOrgCode, Name)
SalesOrganization	(SalesOrgCode, Name, Currency, CompanyCode ^{FK})
DistributionChannel	(DistChannelCode, Name)
Division	(DivisionCode, Name)

Relationships

- Plant *belongs to* CompanyCode (N:1).
- StorageLocation *belongs to* Plant (N:1).
- SalesOrganization *belongs to* CompanyCode (N:1).
- SalesOrganization *serves* Divisions *through* DistributionChannels (ternary M:N:P).

Tasks

Table of Contents

1 Relational Databases – Keys

- Exercise 1A – University Schema
- Exercise 1B – Airline Schema
- Exercise 1C – Event Schema
- Exercise 1D – Music Schema

2 Relational Algebra

- Exercise 2A – Customer / Supplier / Good (Basic)
- Exercise 2B – Customer / Supplier / Good (Harder)
- Exercise 2C – PC / Laptop / Printer (Intermediate)
- Exercise 2D – PC / Laptop / Printer (Advanced Algebra)

3 Normal Forms (1NF – BCNF)

- Exercise 3A – Eurovision Song Contest (Worked Example)
- Exercise 3B – Ticket Schema
- Exercise 3C – Library Loan Schema

4 Normal Forms (4NF / 5NF)

- Exercise 4A – Train / Conductor / Mechanic (4NF)
- Exercise 4B – Line / Station / Day (5NF)
- Exercise 4C – Train / Route / Service / CarriageType (4NF)
- Exercise 4D – Line / Driver / Shift / Depot (5NF)

5 Canonical Cover & Synthesis Algorithm

Supermarket Schema

- **Employee**(EmployeeID, FirstName, LastName, Phone, DeptID^{FK}, HireDate)
- **Department**(DeptID, Name, Floor)
- **Brand**(BrandID, Name, Country, LegalForm)
- **Product**(EAN, Name, Brand^{FK}, Price, AlcoholContent, DeptID^{FK})
- **Storage**(StorageID, Floor, Capacity)
- **StorageProduct**(StorageID^{FK}, EAN^{FK}, Quantity)

Music Schema

- **Artist**(ArtistID, Name, Country, DebutYear)
- **Album**(AlbumID, Title, ArtistID^{FK}, Genre, ReleaseYear, RecordLabel)
- **Track**(TrackID, Title, AlbumID^{FK}, Duration, FeaturedArtistID^{FK} *nullable*)

These two schemas are used throughout all SQL exercises (Sections 9–12).

Exercise 9A – Supermarket – Basic SELECT

Tasks

- 1 Show all employee names and phone numbers.
- 2 Show all brand names and their legal forms.
- 3 Show the first 5 products ordered by name (alphabetically).
- 4 List all products with alcohol content > 0 .
- 5 Show distinct EAN codes aliased as barcode.
- 6 Show the first 5 employees (use LIMIT).

Example

```
-- Task 1 example
SELECT FirstName, LastName, Phone
FROM Employee;
```

Difficulty: Simple Time: 10 min

Exercise 9B – Music – Basic SELECT

Tasks

- 1 Show all album names and genres.
- 2 Show the first 3 record labels (distinct, alphabetically).
- 3 Show track titles aliased as song.
- 4 Show distinct countries of artists.
- 5 Show the first 5 artists.

Hints

```
-- Alias a column
SELECT Title AS song FROM Track;

-- Distinct values
SELECT DISTINCT Country FROM Artist;

-- Limit results
SELECT Name FROM Artist LIMIT 5;
```

Exercise 9C – Supermarket – WHERE Clauses

Tasks

- 1 List all brands from Germany.
- 2 List products not stored anywhere (no entry in StorageProduct).
- 3 List employees whose last name starts with 'N'.
- 4 List products with alcohol content between 1% and 20%.
- 5 List brands that are a GmbH in Germany but **not** in Hamburg.

Useful Operators

```
-- Pattern matching
WHERE LastName LIKE 'N%'

-- Range filter
WHERE AlcoholContent BETWEEN 1 AND 20

-- Negation + multiple conditions
WHERE Country = 'Germany'
      AND LegalForm = 'GmbH'
      AND City <> 'Hamburg'
```

Exercise 9D – Music – WHERE Clauses

Tasks

- 1 List all distinct genres in the Album table.
- 2 List songs that have a featured artist (FeaturedArtistID is not NULL).
- 3 List songs with a title starting with 'L'.
- 4 List songs with duration between 2 and 3 minutes (120 and 180 seconds).
- 5 List all artists from the USA.

Hint

```
-- Check for NULL
WHERE FeaturedArtistID IS NOT NULL

-- Duration in seconds
WHERE Duration BETWEEN 120 AND 180
```

Difficulty: Simple Time: 15 min

Exercise 9E – Supermarket – Aggregation

Tasks

- 1 Count the number of **distinct** countries of origin for brands.
- 2 Create price categories using CASE WHEN:
 - Price < 1.00 € ⇒ 'cheap'
 - 1.00–5.00 € ⇒ 'medium'
 - Price > 5.00 € ⇒ 'expensive'
- 3 Show the number of products per country and city, ordered by count descending.
- 4 Show the **average price** per department, excluding departments 1 and 2.
- 5 Show departments with at least **5 distinct barcodes** using HAVING.

Hint for Task 2

```
SELECT Name ,  
       CASE WHEN Price < 1.00 THEN 'cheap'  
            WHEN Price <= 5.00 THEN 'medium'  
            ELSE 'expensive'  
       END AS PriceCategory  
FROM Product;
```

Exercise 9F – Music – Aggregation

Tasks

- 1 Show all distinct countries of record labels (i.e. all countries artists come from).
- 2 Categorize songs by duration using CASE WHEN:
 - $< 120 \text{ s} \Rightarrow \text{'short'}$, $120\text{--}240 \text{ s} \Rightarrow \text{'medium'}$, $> 240 \text{ s} \Rightarrow \text{'long'}$
- 3 Count artists per country, ordered by count descending.
- 4 Show the total duration per album in seconds, excluding the 'Christmas' genre.
- 5 Show artists who have at least **5 songs** using HAVING.

Hint for Task 5

```
SELECT ArtistID, COUNT(*) AS SongCount
FROM Track
    JOIN Album ON Track.AlbumID = Album.AlbumID
GROUP BY ArtistID
HAVING COUNT(*) >= 5;
```


Table of Contents

1 Relational Databases – Keys

- Exercise 1A – University Schema
- Exercise 1B – Airline Schema
- Exercise 1C – Event Schema
- Exercise 1D – Music Schema

2 Relational Algebra

- Exercise 2A – Customer / Supplier / Good (Basic)
- Exercise 2B – Customer / Supplier / Good (Harder)
- Exercise 2C – PC / Laptop / Printer (Intermediate)
- Exercise 2D – PC / Laptop / Printer (Advanced Algebra)

3 Normal Forms (1NF – BCNF)

- Exercise 3A – Eurovision Song Contest (Worked Example)
- Exercise 3B – Ticket Schema
- Exercise 3C – Library Loan Schema

4 Normal Forms (4NF / 5NF)

- Exercise 4A – Train / Conductor / Mechanic (4NF)
- Exercise 4B – Line / Station / Day (5NF)
- Exercise 4C – Train / Route / Service / CarriageType (4NF)
- Exercise 4D – Line / Driver / Shift / Depot (5NF)

5 Canonical Cover & Synthesis Algorithm

Exercise 10A – Music – JOINS (Simple)

Tasks

- 1 List all songs in the 'Pop' genre (JOIN Track with Album).
- 2 List artists who have songs on Taylor Swift albums.
- 3 List all albums from an Australian artist.
- 4 Show each album with its artist name.
- 5 Show track names with album title **and** artist name (three-table join).

Reminder: JOIN syntax

```
-- INNER JOIN
SELECT t.Title, a.Title AS Album
FROM Track t
     JOIN Album a ON t.AlbumID = a.AlbumID;

-- THREE-TABLE JOIN
SELECT t.Title, al.Title, ar.Name
FROM Track t
     JOIN Album al ON t.AlbumID = al.AlbumID
     JOIN Artist ar ON al.ArtistID = ar.ArtistID;
```

Exercise 10B – Music – JOINS (Intermediate)

Tasks

- 1 List all songs **not** in the 'Christmas' genre.
- 2 List artists who have songs **shorter than 3 minutes** (180 s).
- 3 List artists who have **more than 5 albums**.
- 4 Show artists with their **total number of tracks**.
- 5 List albums that have **no tracks** (use LEFT JOIN + IS NULL).

Hint for Task 5 – Finding albums with no tracks

```
SELECT al.Title
FROM Album al
      LEFT JOIN Track t ON al.AlbumID = t.AlbumID
WHERE t.TrackID IS NULL;
```

Difficulty: Intermediate Time: 20 min

Exercise 10C – Supermarket – JOINS

Tasks

- 1 List all products from **France** (join Brand and Product).
- 2 List products that are in storage but have **no department** assigned.
- 3 List **empty storage rooms** (no products stored there).
- 4 List **brands with no products**.
- 5 List employees who work in departments that have at least **3 products from the USA**.

Hint for Task 3

```
-- Empty storage rooms: LEFT JOIN + IS NULL
SELECT s.StorageID, s.Floor
FROM Storage s
     LEFT JOIN StorageProduct sp
           ON s.StorageID = sp.StorageID
WHERE sp.EAN IS NULL;
```

Table of Contents

1 Relational Databases – Keys

- Exercise 1A – University Schema
- Exercise 1B – Airline Schema
- Exercise 1C – Event Schema
- Exercise 1D – Music Schema

2 Relational Algebra

- Exercise 2A – Customer / Supplier / Good (Basic)
- Exercise 2B – Customer / Supplier / Good (Harder)
- Exercise 2C – PC / Laptop / Printer (Intermediate)
- Exercise 2D – PC / Laptop / Printer (Advanced Algebra)

3 Normal Forms (1NF – BCNF)

- Exercise 3A – Eurovision Song Contest (Worked Example)
- Exercise 3B – Ticket Schema
- Exercise 3C – Library Loan Schema

4 Normal Forms (4NF / 5NF)

- Exercise 4A – Train / Conductor / Mechanic (4NF)
- Exercise 4B – Line / Station / Day (5NF)
- Exercise 4C – Train / Route / Service / CarriageType (4NF)
- Exercise 4D – Line / Driver / Shift / Depot (5NF)

5 Canonical Cover & Synthesis Algorithm

Exercise 11A – INSERT – Supermarket

Tasks

- 1 Insert a new department 'Bakery' on floor 2 into the Department table.
- 2 Insert a new employee 'John Smith' who works in the Bakery department (use the ID you just inserted).

Syntax Reminder

```
-- Full insert
INSERT INTO table_name (col1, col2, col3)
VALUES (val1, val2, val3);

-- Insert using LAST_INSERT_ID() for FK
INSERT INTO Employee (FirstName, LastName, DeptID)
VALUES ('John', 'Smith', LAST_INSERT_ID());
```

Difficulty: Simple Time: 10 min

Exercise 11B – UPDATE – Supermarket

Tasks

- 1 Rename the department 'Non-food' to 'Household'.
- 2 Update the price of the product 'Coca Cola Zero' to 1.29.

Syntax Reminder

```
UPDATE table_name  
SET column = new_value  
WHERE condition;
```

Always use WHERE with UPDATE!

Without a WHERE clause, **every row** in the table will be updated. Check your condition carefully before executing.

Difficulty: Simple Time: 10 min

Exercise 11C – DELETE – Supermarket

Tasks

- 1 Delete the department 'Flowers and Gardening'.
- 2 Delete the employee with last name 'Major'.

Syntax Reminder

```
DELETE FROM table_name  
WHERE condition;
```

Referential Integrity Warning

Deleting a department that is referenced by employees will **fail** unless you:

- Delete the employees first (manual cascade).
- Have defined ON DELETE CASCADE on the FK.
- Set employee DeptID to NULL first (ON DELETE SET NULL).

Difficulty: Simple Time: 10 min

Exercise 11D – INSERT – Music

Tasks

- 1 Insert a new **Artist** of your choice into the Artist table.
- 2 Insert one of their **Albums** into the Album table (referencing the artist you just created).
- 3 Insert at least **3 Tracks** from that album into the Track table.

Remember to respect FK constraints

```
-- Step 1: insert artist
INSERT INTO Artist (ArtistID, Name, Country, DebutYear)
VALUES (999, 'My Artist', 'Germany', 2020);

-- Step 2: insert album referencing that artist
INSERT INTO Album (AlbumID, Title, ArtistID, Genre,
                  ReleaseYear, RecordLabel)
VALUES (888, 'My Album', 999, 'Pop', 2022, 'My Label');
```

Difficulty: Simple Time: 15 min

Exercise 11E – UPDATE – Music

Tasks

- 1 Change the genre of album 'Fearless' to 'Country Pop'.
- 2 Change the record label of album 'Red' to 'Republic Records'.
- 3 Revert the change in task 2: set the record label of 'Red' back to its original value.

Useful pattern – check before updating

```
-- Always SELECT first to verify your WHERE clause
SELECT AlbumID, Title, RecordLabel
FROM Album
WHERE Title = 'Red';

-- Then UPDATE
UPDATE Album
SET RecordLabel = 'Republic Records'
WHERE Title = 'Red';
```

Exercise 11F – DELETE – Music

Tasks

- 1 Delete all tracks that **feature** the artist 'Blackpink' (i.e. where FeaturedArtistID refers to Blackpink).
- 2 Delete the artist 'Ice Spice' from the Artist table.
Note: handle FK constraints – you may need to nullify or delete referencing rows first.

Handling FK constraints before DELETE

```
-- Option A: nullify the featured artist reference first
UPDATE Track
SET FeaturedArtistID = NULL
WHERE FeaturedArtistID = (
    SELECT ArtistID FROM Artist WHERE Name = 'Ice Spice'
);

-- Then delete the artist
DELETE FROM Artist WHERE Name = 'Ice Spice';
```

Table of Contents

1 Relational Databases – Keys

- Exercise 1A – University Schema
- Exercise 1B – Airline Schema
- Exercise 1C – Event Schema
- Exercise 1D – Music Schema

2 Relational Algebra

- Exercise 2A – Customer / Supplier / Good (Basic)
- Exercise 2B – Customer / Supplier / Good (Harder)
- Exercise 2C – PC / Laptop / Printer (Intermediate)
- Exercise 2D – PC / Laptop / Printer (Advanced Algebra)

3 Normal Forms (1NF – BCNF)

- Exercise 3A – Eurovision Song Contest (Worked Example)
- Exercise 3B – Ticket Schema
- Exercise 3C – Library Loan Schema

4 Normal Forms (4NF / 5NF)

- Exercise 4A – Train / Conductor / Mechanic (4NF)
- Exercise 4B – Line / Station / Day (5NF)
- Exercise 4C – Train / Route / Service / CarriageType (4NF)
- Exercise 4D – Line / Driver / Shift / Depot (5NF)

5 Canonical Cover & Synthesis Algorithm

Exercise 12A – CREATE / ALTER – Music

Tasks

- 1 Create a new table **Concert**(ConcertID, ArtistID, Venue, City, Date, Capacity).
- 2 Add a **foreign key** constraint from Concert.ArtistID to Artist.ArtistID.
- 3 Add a column Website (VARCHAR(200), nullable) to the Artist table.
- 4 Change the Genre column in Album to VARCHAR(100).
- 5 Drop the Country column from Artist.

Syntax Reminder

```
CREATE TABLE Concert (  
    ConcertID INT PRIMARY KEY AUTO_INCREMENT,  
    ArtistID INT NOT NULL,  
    Venue VARCHAR(200),  
    City VARCHAR(100),  
    Date DATE,  
    Capacity INT
```

Exercise 12B – CREATE / ALTER – Supermarket

Tasks

- 1 Add columns Salary (DECIMAL(10,2)) and HireDate (DATE) to the Employee table.
- 2 Create a **Manager** table: (ManagerID, EmployeeID^{FK}, Salary, StartDate).
- 3 Add a column ManagerID (nullable FK) to the Department table – a department can have one manager.

Syntax Reminder

```
ALTER TABLE Employee
  ADD COLUMN Salary    DECIMAL(10,2),
  ADD COLUMN HireDate DATE;
```

```
CREATE TABLE Manager (
  ManagerID  INT PRIMARY KEY AUTO_INCREMENT,
  EmployeeID INT NOT NULL,
  Salary     DECIMAL(10,2),
  StartDate  DATE,
  CONSTRAINT fk_mgr_emp
```

Exercise 12C – Cinema Management Database

Tasks

- 1 Create a database named `cinema_management`.
- 2 Create table **Movie**(MovieID AUTO_INCREMENT, Title, Genre, Duration, Rating DECIMAL, ReleaseYear).
- 3 Create table **Cinema**(CinemaID, Name, City, Screens INT).
- 4 Create table **Screening**(ScreeningID, MovieID^{FK}, CinemaID^{FK}, ScreeningDate DATE, StartTime TIME, Seats INT, Price DECIMAL).
- 5 Add all appropriate constraints (NOT NULL, CHECK, DEFAULT).

Partial Skeleton

```
CREATE DATABASE cinema_management;  
USE cinema_management;
```

```
CREATE TABLE Movie (  
    MovieID      INT PRIMARY KEY AUTO_INCREMENT,  
    Title        VARCHAR(255) NOT NULL,  
    Genre        VARCHAR(100),  
    Duration     INT CHECK (Duration > 0)
```

Table of Contents

1 Relational Databases – Keys

- Exercise 1A – University Schema
- Exercise 1B – Airline Schema
- Exercise 1C – Event Schema
- Exercise 1D – Music Schema

2 Relational Algebra

- Exercise 2A – Customer / Supplier / Good (Basic)
- Exercise 2B – Customer / Supplier / Good (Harder)
- Exercise 2C – PC / Laptop / Printer (Intermediate)
- Exercise 2D – PC / Laptop / Printer (Advanced Algebra)

3 Normal Forms (1NF – BCNF)

- Exercise 3A – Eurovision Song Contest (Worked Example)
- Exercise 3B – Ticket Schema
- Exercise 3C – Library Loan Schema

4 Normal Forms (4NF / 5NF)

- Exercise 4A – Train / Conductor / Mechanic (4NF)
- Exercise 4B – Line / Station / Day (5NF)
- Exercise 4C – Train / Route / Service / CarriageType (4NF)
- Exercise 4D – Line / Driver / Shift / Depot (5NF)

5 Canonical Cover & Synthesis Algorithm

Exercise 13 – One-Pager Presentation

Instructions

Form groups of **2–3 students**. Each group picks **one** of the topics below and prepares a **5-minute presentation**.

Option A: Concurrency Problems

Explain **Lost Update**, **Dirty Read**, and **Phantom Read** with concrete, realistic examples. What transaction isolation level prevents each problem?

Option B: CAP Theorem

Explain the three properties (**Consistency**, **Availability**, **Partition Tolerance**) and the trade-offs. Give real-world database examples for each combination (CA, CP, AP).

Option C: BASE vs. ACID

Compare the two consistency models. Explain when each is appropriate. Give concrete examples of systems that use ACID and systems that use BASE.

Exercise 13 – Requirements

Deliverables

- ① **One-pager:** maximum one A4 page to serve as handout / visual aid for classmates.
 - Must be self-contained: someone who hasn't studied the topic should understand the key ideas.
 - Include at least one concrete example.
- ② **Presentation:** exactly 5 minutes (use a timer).
 - All group members must speak.
 - Explain the concept, not just list definitions.

Evaluation Criteria

- Correctness and completeness of the content.
- Clarity of explanation to peers.
- Quality and usefulness of the one-pager.
- Adherence to the 5-minute time limit.

Preparation time: 20 minutes Presentations: in class